Description of the Fair Subspace Benchmark Experiment

P765 Final Project

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1 Purpose

This experiment compares three different basis-state families for quantum subspace methods:

- Krylov basis states,
- UCCSD basis states,
- fixed-architecture hardware-efficient ansatz (HEA) basis states.

The goal is not to compare them as single-state VQE trial functions. Instead, the goal is to compare them fairly as *subspace bases*. Each family is used to generate a small set of states, and those states are then used to build a projected Hamiltonian problem whose lowest eigenvalue approximates the ground-state energy.

2 Molecular Systems and Hamiltonian

The workflow supports the molecules

$$\mathrm{H}_2, \quad \mathrm{H}_4, \quad \mathrm{LiH},$$

using PennyLane quantum chemistry Hamiltonians in the `sto-3g` basis by default. Molecular coordinates are interpreted in `angstrom` units to avoid ambiguity.

For each molecule, the notebook builds:

- the qubit Hamiltonian $\hat{H}$,
- the number of qubits,
- the number of electrons,
- the Hartree–Fock (HF) reference bitstring.

An exact FCI reference energy is also computed by diagonalizing the Hamiltonian in the fixed-electron sector. This gives a common target energy E_{FCI} for all basis families.

3 Basis Families

Krylov basis

Each Krylov state is generated from the HF reference using Trotterized real-time evolution:

$$|\phi(t)\rangle \approx e^{-it\hat{H}}|\phi_{\mathrm{HF}}\rangle.$$

One scalar time parameter is optimized for each added basis state. The Trotter order and number of Trotter steps are fixed globally so that the architecture is the same throughout the benchmark.

UCCSD basis

Each UCCSD basis state starts from the same HF reference and applies PennyLane's `qml.UCCSD` template. The excitation pool is built once from the selected molecule using the electron number and qubit count. For each added basis state, one full parameter vector $\boldsymbol{\theta}$ is optimized.

Fixed-architecture HEA basis

Each HEA basis state also starts from the HF reference. The circuit architecture is fixed globally before optimization:

- number of layers,
- rotation-gate pattern,
- entangler type,
- entangler connectivity pattern.

Only the angle vector is optimized. This makes the HEA comparison fairer, because the HEA is not allowed to change both its circuit structure and its parameters at the same time.

4 Common Subspace Objective

Suppose the benchmark has generated K basis states

$$\{|\phi_0\rangle, |\phi_1\rangle, \ldots, |\phi_{K-1}\rangle\}.$$

For every family, the same projected matrices are constructed:

$$H_{ij} = \langle\phi_i|\hat{H}|\phi_j\rangle, \qquad S_{ij} = \langle\phi_i|\phi_j\rangle.$$

The projected ground-state estimate is obtained from the generalized eigenvalue problem

$$H\mathbf{c} = E_{\mathrm{sub}} S\mathbf{c}.$$

In implementation, the notebook orthogonalizes the basis using the overlap matrix S and solves the equivalent standard eigenvalue problem in the retained subspace.

To avoid unfairly rewarding nearly duplicate basis states, the same conditioning rule is used for all families. If the overlap matrix has eigenvalues below a shared threshold,

$$\lambda_{\min}(S) \leq 10^{-8},$$

the candidate state is treated as invalid and receives a large penalty. This prevents linear dependence from artificially enlarging the basis.

5 Fairness Protocol

The benchmark uses the same rules for Krylov, UCCSD, and HEA:

- the same molecular Hamiltonian,
- the same HF reference state,
- the same basis size target K,
- the same projected-subspace loss function,
- the same overlap-conditioning rule,
- the same optimizer family and optimization budget.

The primary fairness axis is the *number of basis states* K. Total parameter count is reported only as a secondary diagnostic.

The basis is built greedily:

1. Set the first basis state to the HF reference.
2. Optimize the next basis state while freezing all previously chosen states.
3. Recompute the projected energy.
4. Repeat until the target basis size is reached.

The optimizer used in the notebook is `scipy.optimize.minimize` with Powell's method, together with the same restart count and stopping budget for all three families.

6 Default Benchmark Configuration

The default notebook configuration is chosen for fast and reliable execution:

- molecule: H_2,
- bond length: 0.74 Å,
- basis: `sto-3g`,
- unit: `angstrom`,
- maximum basis size: $K_{\max} = 2$.

The choice $K_{\max} = 2$ is intentional for the default H_2 experiment. Starting from the HF reference, the Krylov family can saturate quickly for this small system, so a larger default K may fail to produce another linearly independent Krylov basis state. The notebook still supports larger systems such as H_4 and LiH, where richer subspaces are possible.

7 Default H_2 Results

For the validated default run, the FCI reference energy is

$$E_{\mathrm{FCI}} = -1.137283834704 \text{ Ha}.$$

The projected subspace energies are:

Family	K	E_{sub} (Ha)	$E_{\mathrm{sub}} - E_{\mathrm{FCI}}$ (Ha)
Krylov	1	-1.116759307528	$+0.020524527176$
Krylov	2	-1.137283834704	0.000000000000
UCCSD	1	-1.116759307528	$+0.020524527176$
UCCSD	2	-1.137283794326	$+0.000000040379$
Fixed HEA	1	-1.116759307528	$+0.020524527176$
Fixed HEA	2	-1.117134883583	$+0.020148951121$

These results show that, under the same subspace protocol and at the same basis size:

- Krylov reaches the FCI energy at $K = 2$ for the default H_2 case,
- UCCSD also reaches essentially the same energy at $K = 2$,
- the fixed HEA improves only slightly at $K = 2$ under the same budget.

8 Interpretation

This experiment is designed to answer the following question:

> Given the same Hamiltonian, the same HF starting point, the same subspace size, and the same optimization budget, which basis-state family gives the best projected ground-state energy?

Because the experiment compares *subspaces* rather than single trial states, it is better suited to the final project goal of subspace diagonalization. The benchmark separates two ideas:

- **expressivity at fixed basis size**, measured by the projected energy error,
- **cost or flexibility**, measured secondarily through total trainable parameter count.

In this sense, the notebook provides a controlled and fair comparison of Krylov, UCCSD, and fixed-architecture HEA bases for subspace methods.

9 Implementation Files

The experiment is implemented in:

- `Fair_Subspace_Benchmark.ipynb`
- `Krylov_Basis_States.ipynb`
- `UCCSD_Basis_States.ipynb`
- `HEA_Basis_States.ipynb`